

Chapter 2 — Variables & Data Types

Original educational notes prepared as sample source material for the RAG assignment.

Variables

A variable is a name that refers to a value. Assignment uses the equals sign, such as `age = 20`. The value can later be replaced with another value.

Common Data Types

Important Python data types include `int` for whole numbers, `float` for decimal numbers, `str` for text, `bool` for True or False values, `list` for ordered mutable collections, `tuple` for ordered immutable collections, `dict` for key-value mappings, and `set` for collections of unique values.

Lists and Tuples

A list is mutable, so its elements can be changed after creation. A tuple is immutable, so its elements cannot be changed after creation. Lists use square brackets, while tuples commonly use parentheses.

Type Conversion

Python provides functions such as `int()`, `float()`, `str()`, and `bool()` for converting values between compatible types. Conversion should be used carefully because not every value can be converted successfully.